ASSIGNMENT NO.5: Aggregate Functions and Data Analysis

Questions:

Q1) Create a MySQL database named CollegeDB and a table named StudentResult.

The table should contain the following columns:

Column Name	Data Type	Constraint
StudentID	INT	PRIMARY KEY
StudentName	VARCHAR(50)	
Course	VARCHAR(30)	
City	VARCHAR(30)	
Marks	INT	
FeesPaid	DECIMAL(10,2)	

Insert at least 10 records into the table.

QUERY:

```sql
CREATE DATABASE SY_CollegeDB;
USE SY_CollegeDB;

CREATE TABLE StudentResult (
    StudentID INT PRIMARY KEY,
    StudentName VARCHAR(50),
    Course VARCHAR(30),
    City VARCHAR(30),
    Marks INT,
    FeesPaid DECIMAL(10,2)
);

INSERT INTO StudentResult VALUES
(1,'Aarav Shah','BSc IT','Mumbai',85,75000),
(2,'Meera Joshi','BCA','Pune',78,70000),
(3,'Rohan Patil','BSc CS','Nashik',88,72000),
(4,'Sneha Desai','BCA','Mumbai',92,70000),
(5,'Amit Kumar','BSc IT','Delhi',75,75000),
(6,'Pooja Singh','BSc CS','Pune',81,72000),
(7,'Vikram Rao','BCA','Bengaluru',89,70000),
(8,'Neha Sharma','BSc IT','Nashik',94,75000),
(9,'Rahul Verma','BSc CS','Mumbai',73,72000),
(10,'Kavita Iyer','BCA','Chennai',87,70000);

SELECT * FROM StudentResult;
```

OUTPUT:

←T→	StudentID	StudentName	Course	City	Marks	FeesPaid
Edit Copy Delete	1	Aarav Shah	BSc IT	Mumbai	85	75000.00
Edit Copy Delete	2	Meera Joshi	BCA	Pune	78	70000.00
Edit Copy Delete	3	Rohan Patil	BSc CS	Nashik	88	72000.00
Edit Copy Delete	4	Sneha Desai	BCA	Mumbai	92	70000.00
Edit Copy Delete	5	Amit Kumar	BSc IT	Delhi	75	75000.00
Edit Copy Delete	6	Pooja Singh	BSc CS	Pune	81	72000.00
Edit Copy Delete	7	Vikram Rao	BCA	Bengaluru	89	70000.00
Edit Copy Delete	8	Neha Sharma	BSc IT	Nashik	94	75000.00
Edit Copy Delete	9	Rahul Verma	BSc CS	Mumbai	73	72000.00
Edit Copy Delete	10	Kavita Iyer	BCA	Chennai	87	70000.00

Q2) Write MySQL queries to find the following from the StudentResult table:

Requirement
Total number of students
Total fees paid
Average marks
Lowest marks
Highest marks

QUERY:

```sql
SELECT
    COUNT(*) AS TotalStudents,
    SUM(FeesPaid) AS TotalFeesPaid,
    AVG(Marks) AS AverageMarks,
    MIN(Marks) AS LowestMarks,
    MAX(Marks) AS HighestMarks
FROM StudentResult;
```

OUTPUT:

TotalStudents	TotalFeesPaid	AverageMarks	LowestMarks	HighestMarks
10	721000.00	84.2000	73	94

Q3) Write a MySQL query to count the number of students in each course.

The output should contain:

Column

Course
TotalStudents

QUERY:

```
SELECT Course, COUNT(*) AS TotalStudents
FROM StudentResult
GROUP BY Course;
```

OUTPUT:

Course	TotalStudents
BCA	4
BSc CS	3
BSc IT	3

Q4) Write a MySQL query to find the average marks of students course-wise.

The output should contain:

Column
Course
AverageMarks

QUERY:

```
SELECT Course, AVG(Marks) AS AverageMarks
FROM StudentResult
GROUP BY Course;
```

OUTPUT:

←T→				Course	AverageMarks
☐	Edit	Copy	Delete	BCA	86.5000
☐	Edit	Copy	Delete	BSc CS	80.6667
☐	Edit	Copy	Delete	BSc IT	84.6667

Q5) Write a MySQL query to display only those courses where the number of students is greater than 2.

Use:

Clause
GROUP BY
HAVING

QUERY:

```sql
SELECT Course, COUNT(*) AS TotalStudents
FROM StudentResult
GROUP BY Course
HAVING COUNT(*) > 2;
```

OUTPUT:

Course	TotalStudents
BCA	4
BSc CS	3
BSc IT	3

Q6) Write a MySQL query to analyze course performance.

The output should display:

Column
Course
Column
TotalStudents
AverageMarks
LowestMarks
HighestMarks

Sort the result by `AverageMarks` in descending order.

QUERY:

```sql
SELECT Course,
       COUNT(*) AS TotalStudents,
       AVG(Marks) AS AverageMarks,
       MIN(Marks) AS LowestMarks,
       MAX(Marks) AS HighestMarks
FROM StudentResult
GROUP BY Course
ORDER BY AverageMarks DESC;
```

OUTPUT:

Course	TotalStudents	AverageMarks 1	LowestMarks	HighestMarks
BCA	4	86.5000	78	92
BSc IT	3	84.6667	75	94
BSc CS	3	80.6667	73	88